

Robot that can be controlled with thoughts

According to a new research, it is now possible to control a robotic hand with only your mind. According to Bin He, professor and lead researcher of the study, department of biomedical engineering, University of Minnesota, USA, "This is the first time in the world that people can operate a robotic arm to reach and grasp objects in a complex 3D environment using only their thoughts, without a brain implant."



Research subjects at University of Minnesota fitted with a specialised non-invasive brain cap were able to move the robotic arm just by imagining moving their own arms (Image courtesy: University of Minnesota)

He adds, "Just by imagining moving their arms, people were able to move the robotic arm." This research has the potential to help millions of people who are paralysed or have neurodegenerative diseases.

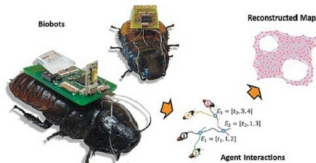
The non-invasive technique, called electroencephalography (EEG) based brain-computer interface, records weak electrical activity of the participant's brain through a specialised, high-tech EEG cap fitted with 64 electrodes and converts the thoughts into action by advanced signal processing and machine learning.

UAVs, biobots for mapping unexplored territories

Researchers at North Carolina State University, USA, have developed a combination of software and hardware that will allow them to use unmanned aerial vehicles (UAVs) and insect cyborgs, or biobots, to map large, unfamiliar areas such as collapsed buildings after a disaster.

Edgar Lobaton, assistant professor of electrical and computer engineering at NC State and co-author of the paper, says, "The idea is to release a swarm of sensor-equipped biobots, like remotely-controlled cockroaches, into a collapsed building or other dangerous, unmapped area."

He adds, "Using remote-control technology, we would restrict the movement of the biobots to a defined area. That area would be defined by proximity to a beacon on a UAV. For example, the biobots may be prevented from going more than 20 metres from the UAV."



The application has utility for areas like collapsed buildings where GPS cannot be used (Image courtesy: <https://news.ncsu.edu>)

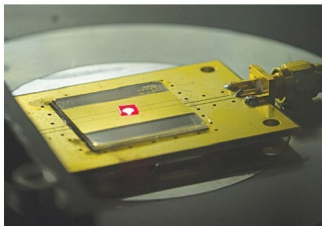
The biobots would be allowed to move freely within a defined area and would signal researchers via radio waves whenever these would get close to each other. Custom software would then use an algorithm to translate biobot sensor data into a rough map of the unknown environment.

Radio receiver built from components the size of two atoms

Researchers from Harvard John A. Paulson School of Engineering and Applied Sciences have made the world's smallest radio receiver, built out of an assembly of atomic-scale defects in pink diamonds.

This tiny radio, whose building blocks are the size of two atoms, can withstand extremely harsh environments and is biocompatible, meaning it could work in places as varied as a probe on Venus to a pacemaker in a human heart.

The diamond's tiny imperfections make up a system of nitrogen atoms with a hole next to it. This arrangement can emit single photons or detect very weak magnetic fields. Also, these flaws have photo-luminescent properties, which means these can convert information into light that could offer uses in sensors and quantum computing.



The radio is extremely resilient, thanks to the inherent strength of the diamond. The team successfully played music at 350°C (Image courtesy: Harvard John A. Paulson School of Engineering and Applied Sciences)